Macroeconomic Country Analysis

Denmark — Norway — Sweden

Penn World Table 11.00 · FRED · Solow Growth Model

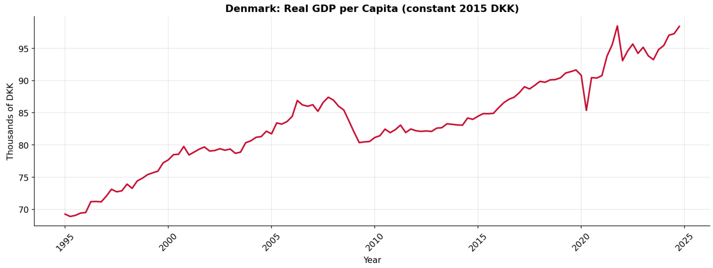
Spring 2026

Denmark

Economic Data & Solow Model

Denmark – GDP per Capita

Trend growth $\approx 1.0\%$ p.a. since 2000 (1995 Q1 – 2024 Q4)

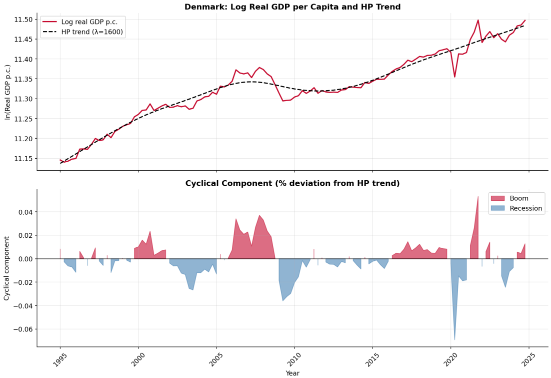


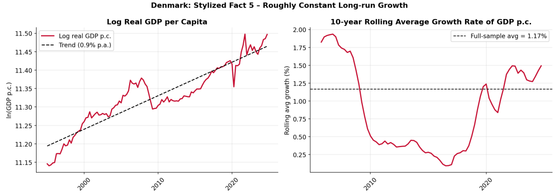
Indicator	Value
Sample	1995–2024
Trend growth	$\approx 1.0\%$ p.a.
GFC drop	$\approx -7\%$
COVID drop	$\approx -5\%$

- Sustained convergence to high-income level
- Clear 2008–09 recession trough
- Rapid post-COVID recovery

Denmark – Business Cycles

HP-filtered output gap ($\lambda = 1600$) · Unemployment & Inflation

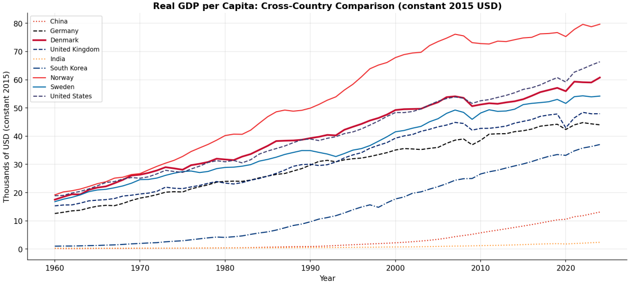


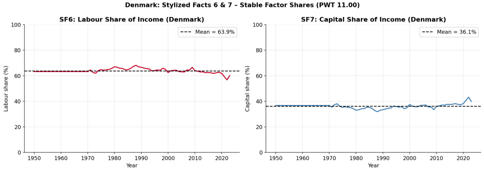


- Output gap autocorrelation $\rho \approx 0.84$ — persistent cycles
- Unemployment $\approx 5\%$ average; CPI trend $\approx 1.8\%$ since 2000
- Strong counter-cyclical unemployment; GFC spike to $\sim 8\%$

Denmark – Stylised Facts

SF1–SF7 from Kaldor; cross-country GDP comparison

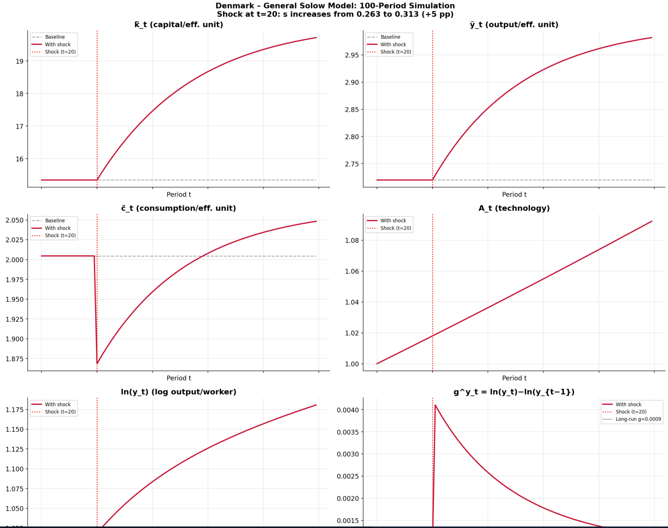




- Broadly confirms SF1–SF7: stable growth, balanced factor shares ($\alpha \approx 0.37$)
- Labour share $\approx 63\%$; capital share relatively stable — consistent with SF6/SF7

Denmark – Solow Model

PWT calibration 2000–2019; savings shock $s\uparrow$



Parameter	Value
α	0.366
s	0.263
δ	0.041
n	0.005
g	0.001
$\tilde{k}^*$	15.35
$\tilde{y}^*$	2.72

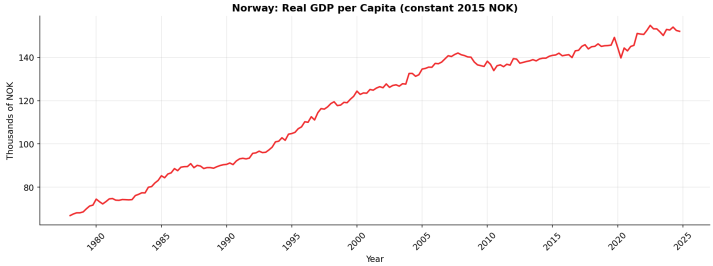
- $s < \alpha$: below Golden Rule
- Savings $\uparrow \Rightarrow \tilde{c}^* \uparrow$ (welfare gain)
- Smooth transition to new SS

Norway

Oil Economy & Solow Anomalies

Norway – GDP per Capita

Trend growth $\approx 1.0\%$ p.a. since 2000 (1978 Q1 – 2024 Q4)

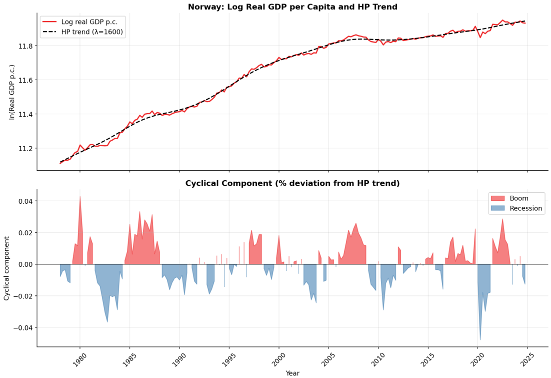


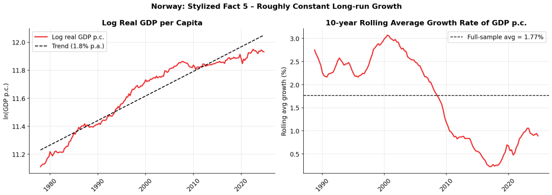
Indicator	**Value**
Sample	1978–2024
Trend growth	$\approx 1.0\%$ p.a.
Oil shock 2015–16	mild dip
COVID drop	$\approx -3\%$

- Highest GDP/capita of three countries
- 1987–89 banking crisis trough
- Oil wealth buffers recessions

Norway – Business Cycles

HP-filtered output gap · lowest output volatility ($\sigma = 0.013$)

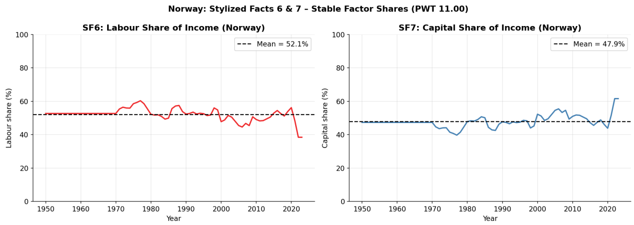


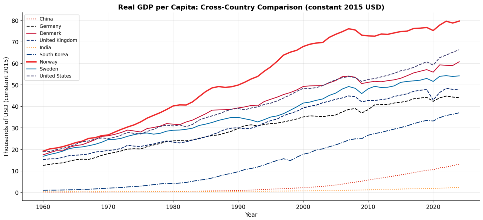


- Output autocorrelation $\rho \approx 0.60$ — moderate persistence
- Unemployment $\approx 3.7\%$ avg since 2000 — lowest of three countries
- CPI $\approx 2.5\%$ since 2000; oil fund stabilises fiscal cycle

Norway – Oil Anomalies

Factor shares violate SF6/SF7; $\alpha = 0.505$ driven by oil rents

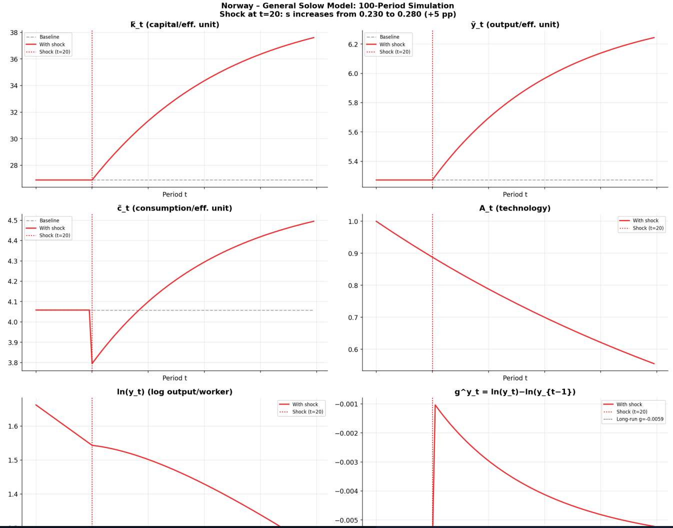




- Labour share swings **38%–65%** with oil price cycle — violates SF6/SF7
- Norway well above peer-group trajectory: oil-wealth premium

Norway – Solow Model

PWT calibration 2000–2019; Oil Fund fiscal rule shock



Parameter	Value
α	0.505
s	0.230
δ	0.042
n	0.009
g	-0.006
$\tilde{k}^*$	26.90
$\tilde{y}^*$	5.27

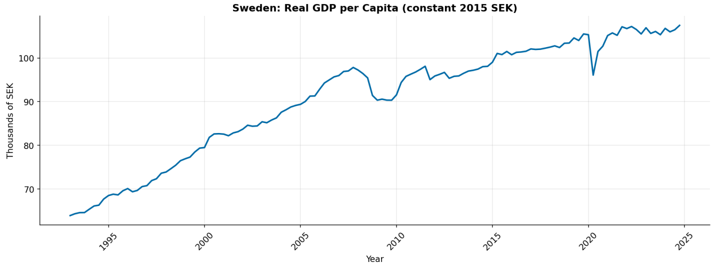
- $g < 0$: maturing oil fields
- $\alpha > 0.5$: oil-inflated capital share
- Shock: reinvest more via GPFG ($s \uparrow$)

Sweden

ERM Crisis Legacy & High Volatility

Sweden – GDP per Capita

Trend growth $\approx 1.25\%$ p.a. since 2000 (1993 Q1 – 2024 Q4)

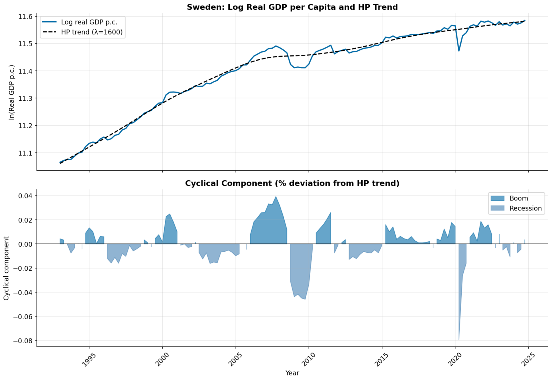


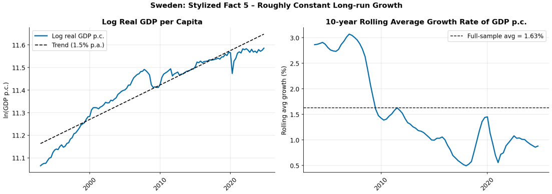
Indicator	**Value**
Sample	1993–2024
Trend growth	$\approx 1.25\%$ p.a.
GFC drop	$\approx -7\%$
COVID drop	$\approx -8\%$

- Fastest-growing of the three countries
- Deepest COVID contraction in sample
- Strong post-GFC V-shaped recovery

Sweden – Business Cycles

Highest output volatility ($\sigma = 0.018$) · Unemployment & Inflation

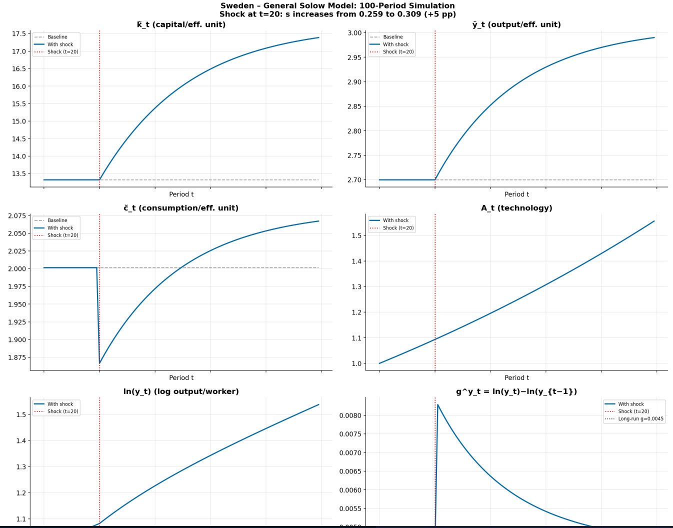




- Output autocorrelation $\rho \approx 0.67$; **highest** output std dev ($\sigma = 0.018$)
- Unemployment peaked $\sim 10\%$ post-1993 ERM crisis, declined to $\sim 6\%$
- CPI $\approx 1.5\%$ since 2000 — below 2% target; Riksbank chronic undershoot

Sweden – Solow Model

PWT calibration 2000–2019; savings shock



Parameter	Value
α	0.384
s	0.259
δ	0.040
n	0.008
g	0.004
$\tilde{k}^*$	13.32
$\tilde{y}^*$	2.70

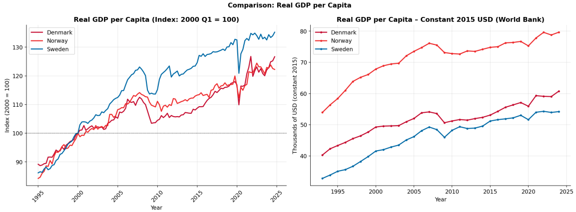
- $s < \alpha$: below Golden Rule
- Savings $\uparrow \Rightarrow \tilde{c}^* \uparrow$ (welfare gain)
- Highest TFP growth of three countries

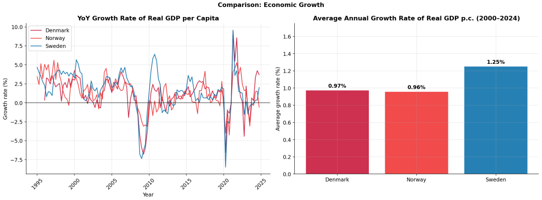
Cross-Country Comparison

Denmark · Norway · Sweden

Comparison – GDP per Capita & Growth

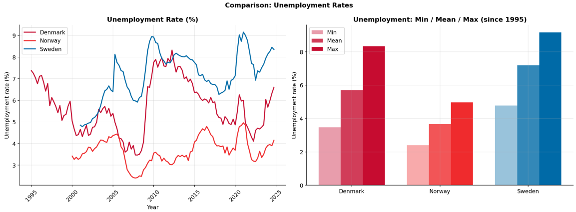
Common sample 1993 Q1 – 2024 Q4

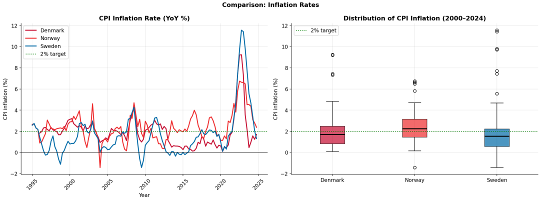




- Norway highest level — oil wealth premium
- Sweden fastest trend growth (≈ 1.25%); Denmark & Norway ≈ 1.0%
- All three show synchronised GFC 2008–09 and COVID 2020 dips

Comparison – Labour Market & Inflation

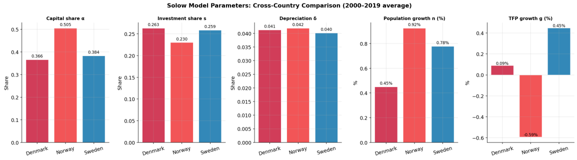


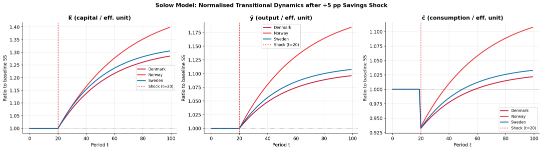


- Norway lowest unemployment (≈ 3.7%); Sweden highest (≈ 6%) — ERM legacy
- Inflation broadly converged since 2000; post-2021 spike common to all three
- Norway oil fund dampens domestic demand swings; Sweden more exposed

Comparison – Solow Model

Cross-country Solow comparison: parameters & steady states





- Norway outlier: $\alpha = 0.505$, $g < 0$ — oil distorts both capital share and TFP
- Denmark & Sweden: similar $\alpha \approx 0.37$, $g > 0$, both below Golden Rule
- All three: $s < \alpha \Rightarrow$ savings increase raises steady-state consumption